

Peter S. Ferguson | CV

Department of Astronomy – University of Washington Seattle
Box 351580, Seattle WA, 98195

 peterferguson.space

 pferguso@uw.edu

 [psferguson](https://github.com/psferguson)

RESEARCH INTERESTS

Observational Cosmology
Photometric Calibration

Milky Way (Sub)structure
Large Scale Surveys

Dark Matter
Instrumentation

Tidal Streams
Science Validation

APPOINTMENTS

Sep. 2024-Present	DiRAC Postdoctoral Fellow, DiRAC institute	U. of Washington Seattle
2021-2024	Postdoctoral Research Associate Rubin Observatory commissioning/Observational cosmology	U. of Wisconsin Madison
2020-2021	Universities Research Association (URA) Visiting Scholar	Fermilab
2016-2021	Graduate Research Assistant: Astronomy & Instrumentation	Texas A&M

EDUCATION

Ph.D. in Astrophysics Texas A&M University Science: Jennifer Marshall & Louis Strigari	Instrumentation: Darren DePoy	2021
B.S. in Astrophysics Haverford College		2013

SCIENTIFIC COLLABORATIONS

2021-Present *Vera Rubin Observatory*: I am an in-kind contributor making significant contributions to many areas of the commissioning process. A few highlights:

- System Integration, Test, and Commissioning (SITCOM): I am one of the core members of the data analysis team helping provide rapid analysis of commissioning efforts (recently the telescope mount and main mirror cells). Additionally, I am helping to develop the tools we will use for the rest of commissioning. I have spent the last 6 months in Chile working closely with the construction team and acting as a volunteer observing specialist.
- Science Verification and Validation (SVV): I have led ad-hoc analyses of Aux-Tel data, been a developer for `analysis_tools` the metric framework for the Rubin Science Pipeline, helped to develop the science validation surveys for commissioning, and am the science lead for the commissioning science unit focused on object detection, quality flags, verification & validation sample production, and survey property maps.
- Data Management (DM): I am one of the core members creating `the_monster` an all sky reference catalog to bootstrap photometric calibrations for early operations. I also can process data and develop code for the rubin science pipelines.

- 2021-Present *Dark Energy Science Collaboration (LSST-DESC)* [<https://lsstdesc.org/>]
Member: Acting as a bridge between the Vera Rubin Observatory project and DESC to assist the Photometric Corrections, Science Release and Validation, and Commissioning working groups.
Dark Matter Working Group Convener: I am one of two conveners for this working group within DESC
- 2018-present *DECam Local Volume Exploration (DELVE) Survey* [<https://delve-survey.github.io>]
Builder: DELVE is a 3-year survey combining archival DECam data with 126 nights of dedicated observing. This survey looks to probe the small scale nature of dark matter by (1) searching for ultra faint MW satellites and stellar streams, (2) studying the satellite population and star formation history around the Large and Small Magellanic Clouds, and (3) deeply imaging around isolated Large Magellanic Cloud analogs to determine their satellite luminosity function. I have contributed to much of the calibration pipeline, data validation, and morphological classifier for our first data release DELVE-DR1 in early 2021.
- 2016-present *Dark Energy Survey (DES)* [<https://www.darkenergysurvey.org>]
- 2020-present *Southern Stellar Stream Spectroscopic Survey (S^5)* [<https://s5collab.github.io/>]

Publications

Orcid: **0000-0001-6957-1627** | ADS Library: **Link**

Mentored students are underlined

Primary Contributor (14)

Arora, Arpit, **Ferguson, P. S.**, Nibauer, Jacob, et al., *No Stream Left Unscathed: The Imprint of a Host Galaxy*, **ApJ**, 2026

Gordon, Yjan A., **Ferguson, P. S.**, Martinez, Michael N., et al., *A Census of Variable Radio Sources at 3 GHz*, **The Open Journal of Astrophysics**, 2026

Boone, K. K., **Ferguson, P. S.**, Tabbutt, M., et al., *Robust Measurement of Stellar Streams around the Milky Way: Correcting Spatially Variable Observational Selection Effects in Optical Imaging Surveys*, **ApJ**, 2026

Cohen, Benjamin, Ji, Alexander P., **Ferguson, P. S.**, et al., *Sifting for a Stream: The Morphology of the 300 S Stellar Stream*, **The Open Journal of Astrophysics**, 2026

Ferguson, P. S., Rykoff, Eli S., Carlin, Jeffrey L., et al., *The Monster: A reference catalog with synthetic ugrizy-band fluxes for the Vera C. Rubin observatory*, **DMTN**, 2025

Chiti, A., Tavangar, K., **Ferguson, P. S.**, et al., *DELVE-ing into the Milky Way's Globular Clusters: Assessing Extratidal Features in NGC 5897, NGC 7492, and Testing Detectability with Deeper Photometry*, **AJ**, 2025

Tsiane, Kabelo, Mau, Sidney, Drlica-Wagner, Alex, ..., **Ferguson, P. S.**, et al., *Predictions for the Detectability of Milky Way Satellite Galaxies and Outer-Halo Star Clusters with the Vera C. Rubin Observatory*, **The Open Journal of Astrophysics**, 2025

Li, Ting S., Ji, Alexander P., Pace, Andrew B., ..., **Ferguson, P. S.**, et al., *S^5 : The Orbital and Chemical Properties of One Dozen Stellar Streams*, **ApJ**, 2022

Ferguson, P. S., Shipp, N., Drlica-Wagner, A., et al., *DELVE-ing into the Jet: A Thin Stellar Stream on a Retrograde Orbit at 30 kpc*, **AJ**, 2022

Drlica-Wagner, A., Ferguson, P. S., Adamów, M., et al., *The DECam Local Volume Exploration Survey Data Release 2*, **ApJS**, 2022

Tavangar, K., Ferguson, P. S., Shipp, N., et al., *From the Fire: A Deeper Look at the Phoenix Stream*, **ApJ**, 2022

Drlica-Wagner, A., Carlin, J. L., Nidever, D. L., Ferguson, P. S., et al., *The DECam Local Volume Exploration Survey: Overview and First Data Release*, **ApJS**, 2021

Ferguson, P. S., Strigari, Louis E., *Three-dimensional structure of the Sagittarius dwarf spheroidal core from RR Lyrae*, **MNRAS**, 2020

Shipp, N., Drlica-Wagner, A., Balbinot, E., Ferguson, P. S., et al., *Stellar Streams Discovered in the Dark Energy Survey*, **ApJ**, 2018

Co-Author (39).....

Anbajagane, D., Chang, C., Drlica-Wagner, A., ..., Ferguson, P. S., et al., *Dark Energy Camera All Data Everywhere cosmic shear project. V. Constraints on cosmology and astrophysics from 270 million galaxies across 13,000 deg² of the sky*, **Phys. Rev. D**, 2026

Atzberger, K. R., Pace, A. B., Kallivayalil, N., ..., Ferguson, P. S., et al., *The DECam MAGIC Survey: Uncovering the Tidal Tails of the Crater II Dwarf Galaxy*, **ApJ**, 2026

Simon, Joshua D., Rodriguez, Joseph E., Yana Galarza, Jhon, ..., Ferguson, P. S., et al., *TOI-7169 b: A Hot Jupiter Transiting a Metal-poor Star*, **AJ**, 2026

To, Chun-Hao, Chang, Chihway, Anbajagane, Dhayaa, ..., Ferguson, P. S., et al., *First detection of weak lensing around spectroscopically confirmed low-mass galaxies*, **Phys. Rev. D**, 2026

Chiti, Anirudh, Placco, Vinicius M., Pace, Andrew B., ..., Ferguson, P. S., et al., *Enrichment by the first stars in a relic dwarf galaxy*, **Nature Astronomy**, 2026

Vera C. Rubin Observatory Team, Acero-Cuellar, Tatiana, Acosta, Emily, ..., Ferguson, P. S., et al., *The Vera C. Rubin Observatory Data Preview 1*, **AJ**, 2026

Oden, Slater J., Nidever, David L., Povick, Joshua, ..., Ferguson, P. S., et al., *Warped and Hooked: Mapping the Magellanic Clouds in Three Dimensions Using Red Clump Stars*, **ApJ**, 2026

Sherman, Nora F., Acevedo, Maria, Brout, Dillon, ..., Ferguson, P. S., et al., *The Dark Energy Bedrock All-sky Supernova Program: Motivation, Design, Implementation, and Preliminary Data Release*, **ApJ**, 2026

Chandler, Colin Orion, Bernardinelli, Pedro H., Jurić, Mario, ..., Ferguson, P. S., et al., *NSF-DOE Vera C. Rubin Observatory Observations of Interstellar Comet 3I/ATLAS (C/2025 N1)*, **ApJ**, 2026

Tan, C. Y., Drlica-Wagner, A., Pace, A. B., ..., Ferguson, P. S., et al., *DELVE Milky Way Satellite Galaxy Census. I. Satellite Population and Survey Selection Function in DES, DELVE, and Pan-STARRS*, **ApJ**, 2026

Tan, C. Y., Cerny, W., Pace, A. B., ..., Ferguson, P. S., et al., *Ultra-faint Milky Way Satellites Discovered in Carina, Phoenix, and Telescopium with DELVE Data Release 3*, **ApJ**, 2026

Acevedo, Maria, Sherman, Nora F., Brout, Dillon, ..., Ferguson, P. S., et al., *The Dark Energy Bedrock All-sky Supernova Program: Cross Calibration, Simulations, and Cosmology Forecasts*, **ApJ**, 2026

Barbosa, F. O., Chiti, A., Limberg, G., ..., **Ferguson, P. S.**, et al., *The DECam MAGIC Survey: A Wide-field Photometric Metallicity Study of the Sculptor Dwarf Spheroidal Galaxy*, **ApJ**, 2025

Choi, Yumi, Olsen, Knut A. G., Carlin, Jeffrey L., ..., **Ferguson, P. S.**, et al., *47 Tuc in Rubin Data Preview 1. Exploring Early LSST Data and Science Potential*, **ApJ**, 2025

Schechter, Paul L., Sluse, Dominique, Zaborowski, Erik A., ..., **Ferguson, P. S.**, et al., *The DELVE Quadruple Quasar Search. I. A Lensed Low-luminosity Active Galactic Nucleus*, **AJ**, 2025

Medoff, Jonah, Mutlu-Pakdil, Burçin, Carlin, Jeffrey L., ..., **Ferguson, P. S.**, et al., *DELVE-DEEP Survey: The Faint Satellite System of NGC 55*, **ApJ**, 2025

Pace, Andrew B., Li, T. S., Ji, A. P., ..., **Ferguson, P. S.**, et al., *Spectroscopic Analysis of Pictor II: a very low metallicity ultra-faint dwarf galaxy bound to the Large Magellanic Cloud*, **The Open Journal of Astrophysics**, 2025

Tan, C. Y., Cerny, W., Drlica-Wagner, A., ..., **Ferguson, P. S.**, et al., *A Pride of Satellites in the Constellation Leo? Discovery of the Leo VI Milky Way Satellite Ultra-faint Dwarf Galaxy with DELVE Early Data Release 3*, **ApJ**, 2025

Cerny, W., Chiti, A., Geha, M., ..., **Ferguson, P. S.**, et al., *Discovery and Spectroscopic Confirmation of Aquarius III: A Low-mass Milky Way Satellite Galaxy*, **ApJ**, 2025

Martinez, Michael N., Gordon, Yjan A., Bechtol, Keith, ..., **Ferguson, P. S.**, et al., *Finding Lensed Radio Sources with the Very Large Array Sky Survey*, **ApJ**, 2025

Teixeira, G., Bom, C. R., Santana-Silva, L., ..., **Ferguson, P. S.**, et al., *Photometric redshifts probability density estimation from recurrent neural networks in the DECam local volume exploration survey data release 2*, **Astronomy and Computing**, 2024

Chiti, Anirudh, Mardini, Mohammad, Limberg, Guilherme, ..., **Ferguson, P. S.**, et al., *Enrichment by extragalactic first stars in the Large Magellanic Cloud*, **Nature Astronomy**, 2024

Usman, Sam A., Ji, Alexander P., Li, Ting S., ..., **Ferguson, P. S.**, et al., *Multiple populations and a CH star found in the 300S globular cluster stellar stream*, **MNRAS**, 2024

McNanna, M., Bechtol, K., Mau, S., ..., **Ferguson, P. S.**, et al., *A Search for Faint Resolved Galaxies Beyond the Milky Way in DES Year 6: A New Faint, Diffuse Dwarf Satellite of NGC 55*, **ApJ**, 2024

Zaborowski, E. A., Drlica-Wagner, A., Ashmead, F., ..., **Ferguson, P. S.**, et al., *Identification of Galaxy-Galaxy Strong Lens Candidates in the DECam Local Volume Exploration Survey Using Machine Learning*, **ApJ**, 2023

Cerny, W., Drlica-Wagner, A., Li, T. S., ..., **Ferguson, P. S.**, et al., *DELVE 6: An Ancient, Ultra-faint Star Cluster on the Outskirts of the Magellanic Clouds*, **ApJ**, 2023

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Gordon, Yjan A., O'Dea, Christopher P., Baum, Stefi A., ..., **Ferguson, P. S.**, et al., *Compact Steep Spectrum Radio Sources with Enhanced Star Formation Are Smaller Than 10 kpc*, **ApJ**, 2023

Cerny, W., Simon, J. D., Li, T. S., ..., **Ferguson, P. S.**, et al., *Pegasus IV: Discovery and Spectroscopic Confirmation of an Ultra-faint Dwarf Galaxy in the Constellation Pegasus*, **ApJ**, 2023

Martínez-Vázquez, C. E., Cerny, W., Vivas, A. K., ..., **Ferguson, P. S.**, et al., *RR Lyrae Stars in the Newly Discovered Ultra-faint Dwarf Galaxy Centaurus I*, **AJ**, 2021

Shipp, Nora, Erkal, Denis, Drlica-Wagner, Alex, ..., **Ferguson, P. S.**, et al., *Measuring the Mass of the Large Magellanic Cloud with Stellar Streams Observed by S⁵*, **ApJ**, 2021

Cerny, W., Pace, A. B., Drlica-Wagner, A., ..., **Ferguson, P. S.**, et al., *Eridanus IV: an Ultra-faint Dwarf Galaxy Candidate Discovered in the DECam Local Volume Exploration Survey*, **ApJ**, 2021

Stringer, K. M., Drlica-Wagner, A., Macri, L., ..., **Ferguson, P. S.**, et al., *Identifying RR Lyrae Variable Stars in Six Years of the Dark Energy Survey*, **ApJ**, 2021

Cerny, W., Pace, A. B., Drlica-Wagner, A., **Ferguson, P. S.**, et al., *Discovery of an Ultra-faint Stellar System near the Magellanic Clouds with the DECam Local Volume Exploration Survey*, **ApJ**, 2021

Hansen, T. T., Riley, A. H., Strigari, L. E., ..., **Ferguson, P. S.**, et al., *A Chemo-dynamical Link between the Gjöll Stream and NGC 3201*, **ApJ**, 2020

Hansen, T. T., Marshall, J. L., Simon, J. D., ..., **Ferguson, P. S.**, et al., *Chemical Analysis of the Ultrafaint Dwarf Galaxy Grus II. Signature of High-mass Stellar Nucleosynthesis*, **ApJ**, 2020

Mau, S., Cerny, W., Pace, A. B., ..., **Ferguson, P. S.**, et al., *Two Ultra-faint Milky Way Stellar Systems Discovered in Early Data from the DECam Local Volume Exploration Survey*, **ApJ**, 2020

Stringer, K. M., Long, J. P., Macri, L. M., ..., **Ferguson, P. S.**, et al., *Identification of RR Lyrae Stars in Multiband, Sparsely Sampled Data from the Dark Energy Survey Using Template Fitting and Random Forest Classification*, **AJ**, 2019

SPIE (7).....

Quint, Bruno C., Daruich, Felipe, Kubánek, Petr, ..., **Ferguson, P. S.**, et al., *Rubin M1M3 support system dynamic performance*, **SPIE**, 2024

Rodeghiero, Gabriele, Rosignoli, Luca, Canestrari, Rodolfo, ..., **Ferguson, P. S.**, et al., *The Vera C. Rubin's M2 support system integration and verification at the TMA*, **SPIE**, 2024

Daruich, Felipe, Aguilar, Christian, Andrew, John, ..., **Ferguson, P. S.**, et al., *Rubin Observatory primary tertiary mirror cell assembly: final integration and commissioning*, **SPIE**, 2024

Stalder, Brian, Munoz, Freddy, Aguilar, Christian, ..., **Ferguson, P. S.**, et al., *Rubin Observatory Simonyi Survey Telescope integrated mount performance*, **SPIE**, 2024

Guy, Leanne P., Bechtol, Keith, Carlin, Jeffrey L., ..., **Ferguson, P. S.**, et al., *Faro: a framework for measuring the scientific performance of petascale Rubin Observatory data products*, **SPIE**, 2022

Ferguson, P. S., Barba, L., DePoy, D. L., et al., *Further development and testing of TCal: a mobile spectrophotometric calibration unit for astronomical imaging systems*, **SPIE**, 2020

Ferguson, P. S., DePoy, D. L., Schmidt, L., et al., *Development of TCal: a mobile spectrophotometric calibration unit for astronomical imaging systems*, **SPIE**, 2018

Unrefereed (4).....

Cerny, William, Pai, Aashay, Drlica-Wagner, Alex, ..., **Ferguson, P. S.**, et al., *Discovery of the Distant, Ultra-faint Milky Way Satellite Aquarius IV with the Vera C. Rubin Observatory Early Data Preview 2*, **Research Notes of the American Astronomical Society**, 2026

Han, Jiwon Jesse, Chiti, Anirudh, Chen, Kai-Feng, ..., **Ferguson, P. S.**, et al., *A Path to an All-Sky Survey with Roman*, **arXiv/2602.21280**, 2026

Carlin, Jeffrey L., **Ferguson, P. S.**, Vivas, A. Katherina, et al., *An Outer-disk SX Phe Variable Star in Rubin Data Preview 1*, **Research Notes of the American Astronomical Society**, 2025

Ferguson, P. S., Shipp, Nora, *The DECam Field of Streams: a deep view of the Milky Way halo*, **arXiv/2506.05469**, 2025

Selected Talks/Posters

Invited.....

- 2026 "*Unveiling Dark Matter with Stellar Streams and Satellite Galaxies in the Era of LSST*",
HEAP Seminar at University of Utah Talk
- 2026 "*Unveiling Dark Matter with the Vera C Rubin Observatory*",
Physics Seminar at Weber State Talk
- 2025 "*Unveiling Dark Matter with Dwarfs and Stellar Streams in the Era of Roman & Rubin*",
Cosmic Cartography With Roman Space Telescope Talk
- 2025 "*Unveiling dark matter in the near-field from present (DES) and future (LSST) cosmological surveys*",
AAS 245 DES-DESC splinter session Talk
- 2024 "*Seeking the nature of dark matter with the Milky Way halo and wide-field photometric surveys*",
Astronomy Colloquium, University of Washington Talk
- 2024 "*LSST Overview*",
Dwarf Galaxies, Stellar Clusters and Streams in the LSST era, Chicago, IL Talk
- 2024 "*Seeking the nature of dark matter with the Milky Way halo and wide-field photometric surveys*",
Cosmology Seminar at Carnegie Mellon University, Pittsburgh, PA Talk
- 2024 "*Seeking the nature of dark matter with the Milky Way halo and wide-field photometric surveys*",
Astronomy Seminar, Dartmouth, NH Talk
- 2024 "*Seeking the nature of dark matter with the Milky Way halo*",
MiFA Colloquium, Minneapolis, MN Talk
- 2023 "*Seeking the nature of dark matter with the Milky Way halo*",
Noirlab South Colloquium, La Serena Chile Talk
- 2023 "*Plenary talk on dark matter working group*", Remote, DESC spring meeting Talk
- 2023 "*Seeking the nature of dark matter with the Milky Way halo*",
UW Madison Astronomy Colloquium Talk
- 2022 "*Calibration of the DELVE survey*", DESC Photometric Corrections Working Group Talk

Contributed.....

- 2026 "*S5 view of the Jet stream*", S5 Collaboration meeting Talk
- 2025 "*Searching for semi-resolved dwarfs in the era of LSST with Synthetic Source Injection (SSI)*",
Galactic Frontiers II Poster
- 2023 "*Dark Matter with Rubin Observatory*", PCW Session organizer
- 2022 "*Calibration of the DELVE survey*", DESC Collaboration Meeting Talk
- 2021 "*The Jet stream in DELVE*"
Streams21: Constraints on Dark Matter, Virtual Talk
- 2020 "*Further development and testing of TCal: a mobile spectrophotometric calibration unit for astronomical imaging systems*", SPIE Astronomical Telescopes + Instrumentation, Virtual Poster
- 2019 "*Exploring the 3D structure of the Sagittarius dSph core using RR Lyrae*"
RRL/Cepheid, Cloudcroft, NM Talk

2018 "*TCal: a mobile spectrophotometric calibration unit for astronomical imaging systems*"
SPIE Astronomical Telescopes + Instrumentation, Austin, Tx.

Poster

PROPOSALS

Below are the successful observing proposals I have been the PI for.

2022 *Probing the Milky Way using DECam and stellar streams*
DECam 2023A (2.5 nights of observations)

2022 *Probing the Milky Way using DECam and stellar streams*
DECam 2022B (3.5 nights of observations)

2019 *Probing the Dynamical Structure of Sagittarius*
VLT/FLAMES cycle 105 (0.5 nights of observations pushed to 2021 due to COVID)

2019 *Probing the Dynamical Structure of Sagittarius*
Gemini south 2020A (18 hours of observations not taken due to COVID)

OBSERVING EXPERIENCE

Rubin Observatory Simonyi Survey Telescope Assisting commissioning and integration efforts	15 nights
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Cerro Tololo Inter-American Observatory Chile – Blanco 4-meter telescope Used DECam both in person and remotely	14 nights
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McDonald Observatory TX, USA – Harlan Smith 2.7-meter telescope Used Tull coude Echelle Spectrograph for R-Process Alliance Observing	20 nights
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Gemini South: Chile – 8-meter telescope Commissioned 2 new filters on FLAMINGOS-2	3 nights
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Gemini North: HI, USA – 8-meter telescope Operated Queue as part of work at Gemini	3 nights
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AWARDS

2021 Spring TSAPS outstanding talk by a graduate student

2020 Fall TSAPS outstanding talk by a graduate student

2020 URA Visiting Scholar at Fermilab award (Sponsor: Alex Drlica-Wagner)

MENTORING

2026-present Emma Lamb-Smith, an astronomy major at UW Seattle, has worked on using machine learning methods to identify new cluster scale strong lenses in LSST data.

2025-present Matthieu Pélissier, a physics PhD student at Université Grenoble Alpes, has worked on measuring the impact of LSST observational systematics on stellar stream density measurements, resulting in a first-author publication.

2024-present Elliott Burdett, a physics major at UW Seattle, has worked on finding new stellar streams in delve data and developing scalable code for stream searches in LSST.

- 2024-present Shriya Prasanna, a physics major at UW Seattle, has worked on modeling stellar streams and fitting for the orbits of their progenitors.
- 2022-2025 Kyle Boone, a physics major at UW Madison, has worked on using synthetic source injection (Balrog) and survey property maps to generate stellar weight maps for DES analyses that resulted in a publication. Kyle is now a grad student at Harvard.
- 2023-2024 Miranda Gorsuch, a physics PhD student at UW Madison, has worked on data analysis for LSST commissioning.
- 2020-2021 Kiyan Tavanagar, an astrophysics major at University of Chicago, has worked on characterizing stellar streams found in DES. Kiyan is currently a graduate student at Columbia.
- 2018-2021 Leo Barba, a physics major at Texas A&M, has worked 3D printing and designing parts for TCal as well as helping to set up and run the instrument. Currently astronomy graduate student at UMass Amherst.
- 2018 Sarah Hughes, an REU student at Texas A&M, helped to design the LabView based software used to run TCal.